

Your Workbench, your data

Lab 1: 40 hands-on minutes on v9

September 2026

Everything you touch today is real **Controlled Tier participant data**. Person-level files never leave the workspace. Printed outputs are aggregate, rounded, and suppression-screened by the scripts — keep them that way. There is no answer key: your numbers are real. This handout gives plausibility checks instead.

0–6 minutes: configuration and preflight

```
git clone https://github.com/jeffreyswigert/aou-statgen-lab.git
cd aou-statgen-lab
cp config.example.sh config.sh
nano config.sh          # fill the values projected on screen
bash scripts/00_preflight.sh
```

Fill `WORKSHOP_BUCKET`, `BILLING_PROJECT`, and `CDR_DATASET` from the values the instructor projects. `config.sh` is git-ignored — it never gets committed. Each script explains its own reasoning in comments written for newcomers: read before you run (`less scripts/02_build_phenotype.sh`; `q` exits), starting with `scripts/common.sh`.

6–11 minutes: stage genotypes, build the phenotype

```
bash scripts/01_fetch_genotypes.sh
bash scripts/02_build_phenotype.sh
```

While the fetch runs (~1–3 minutes), read the two scripts. Then look at what arrived — two lines each, before trusting anything:

```
head -n 2 lab_data/chr22_filtered.fam    # FID convention?
head -n 2 lab_data/chr22_filtered.bim    # ID scheme?
head -n 2 lab_data/ancestry_preds.tsv    # key column name?
```

Checkpoints: samples in the fileset _____; variants _____.

The phenotype build queries height as program source concept **903133** — not the standard LOINC concept, which would silently pool in EHR heights — for a 1-in-10 teaching subset, then reduces to one row per person (median), applies 100–250 cm bounds, and standardizes within sex. Every one of those clauses is a decision; `scripts/build_pheno.py` states each one.

11–16 minutes: phenotype EDA

```
bash scripts/03_explore_phenotype.sh
cat results/aou_pheno_summary.txt
```

The summary is formatted like Stata's `summarize / summarize, detail` — with percentiles where Stata prints Min/Max, because an extreme is one person's exact value. Open `results/aou_pheno_hist.png` and `aou_pheno_kde_sex.png` from the JupyterLab file browser. **Plausibility checks** (not answers): US-cohort sex-specific height means sit near 176 cm (male) and 162 cm (female) with SDs of roughly 7–8 cm; your subset should be in that neighborhood. Funnel counts print rounded to the nearest 100 — explain to your partner why.

16–24 minutes: score the PGI

```
bash scripts/04_build_pgi.sh
cat results/aou_pgi_summary.txt
```

Checkpoints: weight rows matched (variants processed) _____; skipped _____. A very low match rate means a wrong ID scheme or wrong column indices — stop and ask. Open `results/aou_pgi_kde_ancestry`. This one-chromosome PGI is deliberately partial. What matters is that you counted every match and every join on real data.

24–29 minutes: the regression

```
bash scripts/05_pgi_regression.sh
cat results/aou_pgi_regression.txt
```

The model: standardized height on the standardized PGI, adjusting for age, sex, and five genetic PCs (parsed from AoU’s ancestry file — note its stringified-list column, handled in the script). **Checkpoints:** the PGI coefficient _____; its incremental R^2 _____. Read the headline row aloud in words: “one SD higher in this PGI is associated with ... SD of height, conditional on the covariates.” Then read the three honest labels at the bottom of the output.

29–34 minutes: interpret with a partner

1. Three joins, three key spellings — name them.
2. Why the median per person? Name one alternative and its trade-off.
3. Say the two-sentence honest reading of the ancestry figure (distributions *and* accuracy).
4. Why report the PGI’s *incremental* R^2 rather than its R^2 alone? What would removing the PCs do to the coefficient?
5. What one variable changes to run everyone rather than 1-in-10? All 22 chromosomes?

34–40 minutes: save and finish

```
bash scripts/06_save_run.sh
```

In AoU mode the disclosure screen **blocks** the save on small-count findings; today’s tables are designed to pass, and the screen’s report will list your person-level files as workspace-only — read that list. Verify the cloud listing, **stop your cloud app** (VMs bill while idle), and finish the exit check: *I pulled real records, defended every cleaning step, counted every join, and put a real PGI in a model I can defend.*

GitHub is part of the workflow

You are already using the pattern: this folder is a `git clone`, and `git pull` in it fetches updates onto your VM. Run `git log -oneline | head` to see the audit trail of how these materials evolved, and after your save, `grep code_version results/run_manifest.txt` — every archived run records the exact commit that produced it. The rules that make git safe here: version **code, config templates, and docs**; never commit participant-level files, results, filled configs (`config.sh` is git-ignored on purpose), or notebooks with outputs — a saved `.ipynb` embeds its cell outputs, and one push can carry real data to GitHub. `.gitignore` is a compliance tool: make the safe thing the default in your own repos too.

Walk the same steps with another variable

The pipeline is a one-variable switch, driven by the menu in `data/phenotypes.tsv` (program measurements and EHR labs, each with vetted concept IDs, bounds, and caveats):

```
bash scripts/02_build_phenotype.sh list
PHENO=ldl bash scripts/02_build_phenotype.sh
bash scripts/03_explore_phenotype.sh
```

DRY_RUN=1 on step 02 prints the SQL without spending a query. The posted weights are for height, so with another phenotype the step-05 regression is cross-trait — its output will remind you.

Taking this to your own project

Change the concept ID, bounds, and one-row rule for your phenotype; swap in your trait's published weights; set `SAMPLE_MOD=1`; loop `CHROM` over 1–22 and sum the `.sscore` SUM columns. Rehearse in the sandbox first — everything except the BigQuery step tests locally for free.